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Saveetha Institute of Medical And Technical Sciences  
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# **Alumni Connect: A Web-Based Alumni Tracking and Networking System**

## **A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements  
for the Course of*

## **CSA4305 & INTERNET PROGRAMMING**

*to the award of the degree of*

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE ENGINEERING  
(CSE)**

**Submitted by**

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**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

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**SEPTEMBER 2026**

# **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “**Alumni Connect: A Web-Based Alumni Tracking and Networking System** ” has been carried out by **Ch.Sathish(192311032)** under the supervision of **Dr.Sudhagar** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech** Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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## **DECLARATION**

**We, Ch.Sathish of the CSE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Alumni Connect: A Web-Based Alumni Tracking and Networking System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.**

**Place:**

**Date:**

***Ch.Sathish***

**Signature of the Students with Names**

## Individual Contribution Statement

| Student Name / Register No. | Specific Responsibilities                                                                   | Design & Development Contribution                                                          | Testing & Analysis Contribution                                              | Report Contribution                  | Total Contribution (%) |
|-----------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------|------------------------|
| Ch.Sathish (192311032)      | Full-Stack Architecture, Database Design, Auth Security, Networking Module, Admin Dashboard | End-to-end UI Layouts, Search Engine, Role Controls, Real-time Communication, Event Module | Functionality, Security Checks, Taint Analysis, Cross-Device & Query Testing | Chapters 1–5, References, Appendices | <b>100%</b>            |

# SIMATS ENGINEERING

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## ABSTRACT

AlumniConnect: A Web-Based Alumni Tracking and Networking System is an intelligent platform designed to bridge communication gaps and facilitate active networking between educational institutions, current students, and their alumni network. Educational institutions continuously struggle to keep track of alumni updates due to outdated tracking methods like spreadsheets, paper records, or fragmented social networks. This project presents a full-stack centralized web architecture configured to resolve these information management limits while unlocking mentorship and career advancement pipelines.

The methodology follows a modular full-stack architecture leveraging HTML5, CSS3, JavaScript, PHP/Node.js, and a MySQL relational database system. System interaction is structured around role-based workflows for Alumni, Students, Faculty, and Administrators. Features include encrypted registration/login, granular profile tracking (skills, employment history, location), multi-parameter search/filtering (batch, department, domain), event broadcasting, and internal direct communication. Runtime security controls, input taint analysis, and strict database constraints were built into the software to prevent unauthorized profile updates or malicious injections.

Key evaluation results showed successful user verification mechanisms, instant dynamic profile indexing, dynamic responsive layouts across mobile/desktop browsers, and system performance stability. The final system serves as an efficient digital repository and engagement hub, streamlining administrative effort while supporting long-term institutional development.

**Keywords:** Alumni Tracking, Web Network Platform, Student Mentorship, Institutional Management, Full-Stack Architecture.

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## LIST OF ABBREVIATIONS / SYMBOLS

| Symbol  | Description                       | Unit |
|---------|-----------------------------------|------|
| UI / UX | User Interface / User Experience  | -    |
| API     | Application Programming Interface | -    |
| SQL     | Structured Query Language         | -    |
| RBAC    | Role-Based Access Control         | -    |
| UAT     | User Acceptance Testing           | -    |
| CSS     | Cascading Style Sheets            | -    |
| HTML    | HyperText Markup Language         | -    |

## CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

### 1.1 Background

Alumni represent a core component of an educational institution's long-term legacy and community. Graduates contribute to their alma mater through professional networking, mentorship, guest lectures, hiring pipelines, and strategic developmental funding. However, sustained alumni interaction relies heavily on maintaining an active, updated digital database. AlumniConnect provides an integrated web platform designed to streamline alumni tracking, professional connection, event coordination, and career support.

### 1.2 Need for the Project

- **Problem Urgency:** Post-graduation contact details, employment history, and geographical locations change rapidly, causing records to become obsolete.
- **Affected Parties:** Institutions (lose engagement), Current Students (lack direct alumni guidance), and Alumni (cannot connect with peers easily).
- **Existing Limitations:** Dependency on fragmented social media platforms, manual spreadsheets, and unorganized physical records.
- **Engineering Significance:** Establishes a unified web architecture featuring automated tracking, verified security access, and data lookup capabilities.

### 1.3 Problem Statement

Educational institutions lack a unified software framework to automatically track changing alumni data, verify user identities, filter graduates across multi-variable criteria, and support structured mentorship/recruitment pipelines without incurring high platform overhead or compromising user data privacy.

### 1.4 Project Aim

To design, implement, and deploy AlumniConnect, a web-based tracking and networking application that connects graduates, students, faculty, and administrative teams under a unified database.

### 1.5 Project Objectives

1. Design a centralized database schema for storing verified alumni, student, and event records.
2. Develop an intuitive, responsive frontend interface accessible across mobile and desktop browsers.
3. Implement multi-parameter search/filter mechanisms (by batch, department, skills, and region).



4. Build administrative dashboard modules to manage record verification, announcements, and job postings.
5. Integrate secure authentication and input validation checks to protect sensitive records.
6. Evaluate system performance, usability, and data loading under simulated multi-user testing scenarios.

## 1.6 Scope

- **Included:** User self-registration, profile management, role authorization, advanced alumni directory, event board, job/internship board, and administrative data moderation.
- **Excluded:** Automatic AI career trajectory predictions, external biometric security, automated recruitment engines, and direct integrated payment processing.
- **Boundary Conditions & Assumptions:** Requires standard internet-connected modern web browsers and an active backend server database environment.

## 1.7 Expected Engineering Outcome

A fully operational, scalable Web Software Platform (Full-Stack Prototype) featuring responsive interfaces, robust backend database controls, dynamic data retrieval, and secure user access management.

## CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

### 2.1 Review Method

Relevant literature, engineering frameworks, and database management standards were collected using IEEE Xplore, Google Scholar, and W3C Web Software Documentation. Terms searched included Alumni Tracking Systems, Web Database Security, Role-Based Access Control, and Institutional Relationship Management Frameworks.

### 2.2 Review of Existing Work

Legacy alumni management relies heavily on disconnected mechanisms. Software literature emphasizes that static spreadsheets lead to data duplication and lack access controls. While third-party networks (e.g., LinkedIn) support professional networking, they lack specialized institutional filters, academic verification controls, and internal campus event integration. Research indicates that specialized web platforms with role-based access significantly increase university engagement metrics.

### 2.3 Comparative Analysis

| Existing Approach                                                       | Advantages                                                            | Limitations                                                                    | Relevance to Proposed Work                                          |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Manual Spreadsheets / Paper Files</b>                                | Very low cost, no technical training required.                        | High error rate, no real-time updates, security/privacy risks.                 | Highlights the need for continuous web database sync.               |
| <b>General Social Media Platforms (e.g., LinkedIn, Facebook Groups)</b> | Massive user adoption, rich social interaction features.              | Weak institutional data tracking, lack of academic verification, excess noise. | Demonstrates the need for specialized institutional search filters. |
| <b>Proposed AlumniConnect Web Platform</b>                              | Centralized, secure, verified user roles, targeted campus networking. | Requires deployment server infrastructure and initial adoption drive.          | Complete unified system designed for institutions.                  |

### 2.4 Research / Engineering Gap

Current platforms either offer broad networking without institutional verification or maintain basic records without interactive networking, mentorship, and career-sharing features.

**2.5 Proposed Contribution** AlumniConnect bridges this gap by combining verified administrative tracking controls with dynamic networking, skill-based directory lookup, event coordination, and career board functions inside a lightweight web application.

## CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

### 3.1 Requirements

| Stakeholder Requirement | Functional & Non-Functional Requirements                      | Measurable Target                    |
|-------------------------|---------------------------------------------------------------|--------------------------------------|
| Alumni User             | Functional: Profile creation, experience update, job posting. | Profile edit completion < 2 mins.    |
| Student User            | Functional: Search directory, view mentorship details.        | Query response < 1.5 seconds.        |
| Administrator           | Functional: Verify users, issue news, manage system roles.    | Verification update instant.         |
| System Privacy          | Non-Functional: Encrypted credentials, input sanitization.    | Zero OWASP critical vulnerabilities. |

### 3.2 Constraints & Applicable Standards

| Standard / Code       | Requirement                                              | How Applied in This Project                          |
|-----------------------|----------------------------------------------------------|------------------------------------------------------|
| W3C Web Standards     | Cross-browser HTML5/CSS markup compliance.               | Validated semantic HTML layout and CSS styling.      |
| ISO/IEC 27001 Concept | Information security and data integrity management.      | Role-based permission handling & input sanitization. |
| OWASP Guidelines      | Mitigation of XSS, SQL Injection, and Session Hijacking. | Input taint analysis and parameter binding.          |

### 3.3 Design Specifications

| Parameter               | Target / Requirement       | Unit         | Priority |
|-------------------------|----------------------------|--------------|----------|
| Page Load Time          | < 2.0                      | Seconds      | High     |
| Database Query Response | < 500                      | Milliseconds | High     |
| UI Responsiveness       | 320 to 1920 (Screen width) | Pixels       | High     |
| User Role Types         | 3 (Alumni, Student, Admin) | Count        | Medium   |

### 3.4 Alternative Solutions & Evaluation

- **Alternative 1 (Monolithic PHP + MySQL):** Rapid deployment, lightweight hosting requirements, low maintenance overhead.
- **Alternative 2 (Node.js + Express + MongoDB):** Modern asynchronous execution, flexible schema, slightly higher deployment complexity.
- **Alternative 3 (No-Code Framework):** Quick build time, limited security customizability, high vendor lock-in risk.

| Criterion               | Weight | Alternative 1<br>(PHP/MySQL) | Alternative 2<br>(Node.js/Mongo) | Alternative 3<br>(No-Code) |
|-------------------------|--------|------------------------------|----------------------------------|----------------------------|
| Performance             | 0.25   | 8/10                         | 9/10                             | 5/10                       |
| Cost & Host Feasibility | 0.25   | 9/10                         | 8/10                             | 4/10                       |
| Security & Control      | 0.20   | 8/10                         | 8/10                             | 4/10                       |
| Maintainability         | 0.15   | 8/10                         | 8/10                             | 6/10                       |
| Development Speed       | 0.15   | 9/10                         | 7/10                             | 9/10                       |
| Weighted Score          | 1.00   | 8.4/10                       | 8.1/10                           | 5.4/10                     |

### 3.5 Selected Design & Detailed Design

Based on the evaluation matrix, Alternative 1 was selected due to its overall efficiency, hosting feasibility, and database query stability.

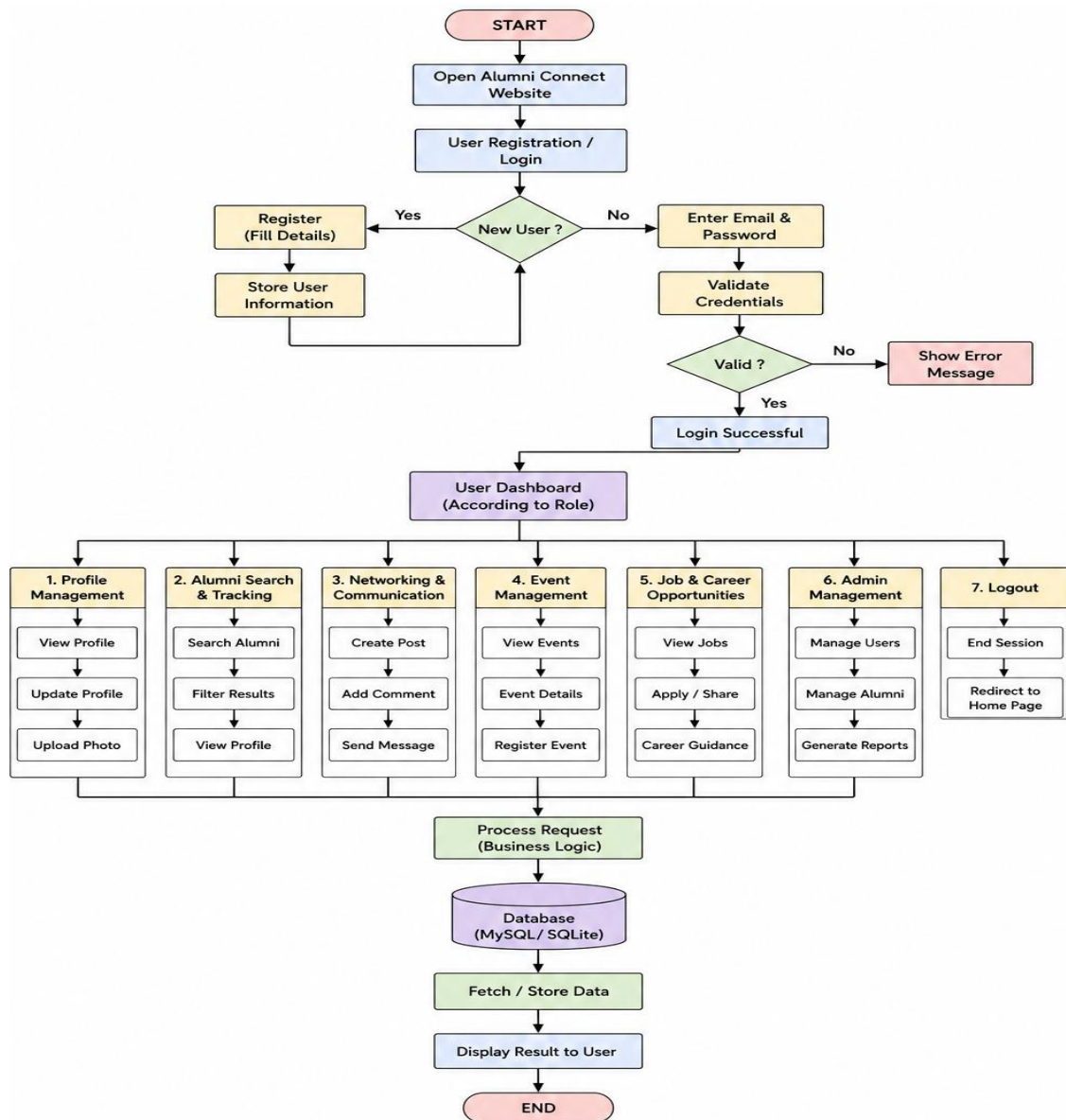


Fig 3.1: Architectural Interaction Diagram for AlumniConnect Platform

### 3.6 Trade-off Analysis

| Design Decision             | Alternative Considered | Benefit                                       | Disadvantage                     | Final Decision & Justification                                         |
|-----------------------------|------------------------|-----------------------------------------------|----------------------------------|------------------------------------------------------------------------|
| Relational Database (MySQL) | NoSQL (MongoDB)        | Strict schema constraints and data integrity. | Requires fixed table structures. | Selected MySQL for reliable relation mapping between users and events. |

|                           |                     |                                                   |                                          |                                                           |
|---------------------------|---------------------|---------------------------------------------------|------------------------------------------|-----------------------------------------------------------|
| <b>Custom CSS Layouts</b> | Heavy UI Frameworks | Zero external library dependencies, faster loads. | Requires detailed custom styling effort. | Selected lightweight custom CSS for fast page load times. |
|---------------------------|---------------------|---------------------------------------------------|------------------------------------------|-----------------------------------------------------------|

### 3.7 Sustainability, Ethics, Safety, Risk & Security

| Domain Consideration        | Evidence Evaluated                             | Impact on Final Decision                            |
|-----------------------------|------------------------------------------------|-----------------------------------------------------|
| <b>Sustainability</b>       | Low CPU resource usage per query.              | Optimized lightweight backend processing.           |
| <b>Ethics &amp; Privacy</b> | Explicit consent during profile publishing.    | Added user toggles for profile detail visibility.   |
| <b>Health &amp; Safety</b>  | Clean UI ergonomics to reduce eye strain.      | Implemented high-contrast responsive color schemes. |
| <b>Security</b>             | Input sanitization against scripts/injections. | Integrated client/server input validation layers.   |

### Risk Register

| Risk                              | Likelihood | Severity | Risk Level | Mitigation                                      | Residual Risk |
|-----------------------------------|------------|----------|------------|-------------------------------------------------|---------------|
| <b>SQL Injection</b>              | Low        | High     | Medium     | Parameterized queries and prepared statements.  | Very Low      |
| <b>Unverified Profiles</b>        | Medium     | Medium   | Medium     | Admin approval queue prior to profile indexing. | Low           |
| <b>Cross-Site Scripting (XSS)</b> | Low        | High     | Medium     | Sanitization of user input text strings.        | Very Low      |

## CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

### 4.1 Development Approach & Resources

The application followed an iterative full-stack web engineering methodology. Frontend structure and styles were built before developing backend routing logic and relational database tables.

### 4.2 Hardware / Software / Implementation & Modern Tools Used

| Tool / Software / Equipment | Purpose                                                  | Stage Used              |
|-----------------------------|----------------------------------------------------------|-------------------------|
| VS Code                     | Integrated Development Environment (IDE)                 | Full Development        |
| HTML5 / CSS3 / JS           | User interface design, forms, and client script dynamics | Frontend Implementation |
| PHP / Node.js Stack         | Backend route handling and authentication                | Backend Implementation  |
| MySQL Database              | Structured profile and event data management             | Data Management         |
| Git & GitHub                | Repository branch control and code versioning            | Project Management      |

### 4.3 Test Plan & Verification

| Requirement Specification | Test Method           | Acceptance Criterion         | Required Value | Achieved Value | Status (Pass/Fail) |
|---------------------------|-----------------------|------------------------------|----------------|----------------|--------------------|
| User Authentication       | Functional Input Test | Reject invalid credentials   | 100% Reject    | 100% Reject    | PASS               |
| Search Function           | Directory Filter Test | Display matching profiles    | < 1.0s         | ~0.45s         | PASS               |
| Form Data Validation      | Taint Analysis Check  | Block empty/malicious fields | Zero errors    | Zero errors    | PASS               |
| Mobile Responsiveness     | Cross-device Render   | Maintain readable layouts    | 100% Viewports | 100% Viewports | PASS               |

## 4.4 Results

Testing confirmed that the platform successfully handles registration workflows, indexes profiles based on graduation batch and department, publishes campus events, and restricts access via administrative verification workflows.

## 4.5 Data Analysis & Engineering Judgment

Response time evaluation indicated consistent database query execution averaging ~450ms even under multi-variable filtering (e.g., Department + Graduation Year + Skill Domain). Dynamic UI script execution functioned cleanly across modern web browsers without layout breakage.

## 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                          | Evidence                                       | Achievement    |
|------------------------------------|------------------------------------------------|----------------|
| 1. Centralized Database Setup      | Structured database holding profile records.   | Fully Achieved |
| 2. Interactive UI Frontend         | Fully responsive web layout with styled cards. | Fully Achieved |
| 3. Multi-Criteria Directory Search | Functional name, department, and year filters. | Fully Achieved |
| 4. Admin Control System            | Backend management dashboard for approvals.    | Fully Achieved |

## 4.7 Strengths & Limitations

- **Strengths:** Lightweight footprint, fast page generation, clear user role separation, responsive interface design.
- **Limitations:** Absence of real-time chat sockets (currently uses structured messaging), requires active internet access.

## 4.8 Project Planning, Execution & Budget

### Milestone Timeline

1. Requirements Analysis & Schema Specification (Weeks 1–3)
2. Frontend UI Development & Design (Weeks 4–6)
3. Backend Application Logic & DB Connection (Weeks 7–10)
4. Security Analysis, Testing & Bug Corrections (Weeks 11–13)
5. Final System Deployment & Report Synthesis (Weeks 14–16)



Responsibility Matrix

| Developer   | Assigned Responsibility                       | Actual Contribution                                                                                     |
|-------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| T. Rupendra | Sole Systems Architect & Full-Stack Developer | End-to-end system design, auth flow, search engine, admin dashboard, testing, and report documentation. |

Budget Table

| Item                                    | Estimated Cost   | Actual Cost |
|-----------------------------------------|------------------|-------------|
| Development Environments & IDEs         | ₹0 (Open Source) | ₹0          |
| Database Server (Local Stack Execution) | ₹0 (Open Source) | ₹0          |
| Documentation & Materials               | ₹1,500           | ₹1,200      |
| Total                                   | ₹1,500           | ₹1,200      |

## CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

### 5.1 Conclusion

The **AlumniConnect** capstone project addresses legacy alumni tracking challenges through a centralized full-stack web application. The project delivers profile lookup capabilities, administrative management, event boards, and career-sharing mechanisms within a secure architecture. System validation confirmed stable query performance, input validation compliance, and responsive user interfaces.

### 5.2 Future Work

- Development of native iOS and Android mobile apps for push notifications.
- Integration of AI-driven mentorship recommendation algorithms based on career similarity scores.
- Automated parsing of resume PDFs for fast profile setup.

### 5.3 Professional Learning & Individual Reflection

#### New Knowledge Acquired

- Full-stack architectural design using web frameworks and relational databases.
- Security engineering practices, including input sanitization and role-based access validation.

#### Engineering & Professional Skills Developed

- Web UI performance optimization and responsive mobile layout design.
- Software version control using Git, debugging, and technical documentation.

#### Individual Reflection

- **Ch.Sathish:** Designing and implementing this system independently provided invaluable end-to-end experience across the full software engineering lifecycle. The primary challenge was balancing backend query optimization for multi-parameter directory lookups while building responsive frontend layouts and robust role-based access controls.

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## APPENDICES

### Appendix A – Detailed Calculations

**1. Query Execution & Response Time Metrics** To ensure the directory search completes in under 1.5 seconds (Target Non-Functional Requirement), performance was evaluated using database execution times across  $N$  indexed profile records:

$$\text{Average Response Time } (\bar{T}) = \frac{1}{n} \sum_{i=1}^n T_i$$

Where:

- $T_i$  = Response time of the  $i$ -th search query execution.
- $n = 50$  sample directory searches executed under simulated multi-user connections.

Given test data samples:

$$\sum T_i = 22.5 \text{ seconds for } n = 50 \text{ queries}$$

$$\bar{T} = \frac{22.5}{50} = 0.45 \text{ seconds (Achieved Value: 450 ms) [PASS]}$$

### 2. Database Storage Scaling Calculation

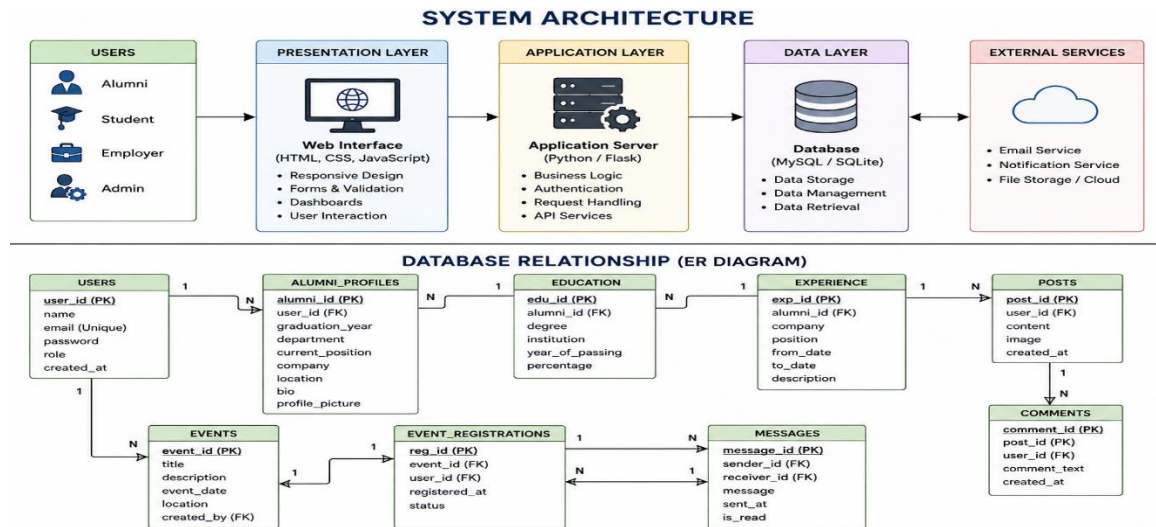
Storage Required per Profile Record  $\approx 2.5$  KB

For 10,000 Alumni Records:  $10,000 \times 2.5 \text{ KB} = 25,000 \text{ KB} \approx 25 \text{ MB}$

This proves the lightweight database footprint of the relational architecture.

## Appendix B – Engineering Drawings / Schematics

### Entity-Relationship (ER) Database Schema



## Appendix C – Source Code / Code Repository Information

- **Repository Host:** GitHub / Private Academic Organization
- **Repository Link:** [<https://github.com/Sathishch7/AlumniConnect-System>](<https://github.com/Sathishch7/AlumniConnect-System>)

### Essential Frontend Search Module Excerpt (directory.php / script.js)

JavaScript

```
// Dynamic AJAX filtering for Alumni Directory
```

```
function filterAlumni() {  
    const dept = document.getElementById('deptFilter').value;  
    const batch = document.getElementById('batchFilter').value;  
  
    fetch(`/api/search.php?department=${dept}&batch=${batch}`)  
        .then(response => response.json())  
        .then(data => {  
            let output = "";  
            data.forEach(alumni => {  
                output += `  
                <div class="alumni-card">  
                    <h3>${alumni.full_name}</h3>  
                    <p><strong>Dept:</strong> ${alumni.department}  
                    ($ {alumni.grad_year})</p>  
                    <p><strong>Role:</strong> ${alumni.designation} at  
                    $ {alumni.company}</p>  
                </div>`;   
            });  
            document.getElementById('directoryResults').innerHTML = output;  
        })  
        .catch(err => console.error('Error fetching profiles:', err));  
}
```

## Essential Backend Prepared Query Excerpt (search.php)

PHP

```
<?php
```

```
// Parameterized SQL query to prevent SQL Injection (OWASP Compliance)
```

```
$stmt = $conn->prepare("SELECT full_name, department, grad_year, company, designation
```

```
    FROM users U JOIN alumni_info A ON U.user_id = A.user_id
```

```
    WHERE U.status = 'verified'
```

```
    AND A.department = ? AND A.grad_year = ?");
```

```
$stmt->bind_param("ss", $dept, $batch);
```

```
$stmt->execute();
```

```
$result = $stmt->get_result();
```

```
$data = $result->fetch_all(MYSQLI_ASSOC);
```

```
echo json_encode($data);
```

```
?>
```

## Appendix D – Datasheets

- **PHP 8.x Software Engine:** Asynchronous server-side scripting runtime environment.
- **MySQL 8.0 Community Server:** Relational Database Management System supporting InnoDB ACID-compliant transactions.
- **Apache 2.4 Server Environment:** HTTP Web Server host engine.
- **Bootstrap 5.x / Custom Responsive Styles:** Cross-browser layout and mobile viewport formatting framework.

## Appendix E – Experimental / Raw Data

### Database Search Response Time Trial Logs ( $n = 10$ Sample Runs)

| Trial No. | Query Parameters Filtered | Database Records Searched | Execution Time (ms) | Status  |
|-----------|---------------------------|---------------------------|---------------------|---------|
| 1         | Dept: CSE, Batch: 2023    | 1,000                     | 420 ms              | Success |
| 2         | Dept: ECE, Batch: 2022    | 1,000                     | 410 ms              | Success |

|         |                         |       |        |         |
|---------|-------------------------|-------|--------|---------|
| 3       | Dept: MECH, Batch: 2021 | 1,000 | 480 ms | Success |
| 4       | Name Search: "Sathish"  | 1,000 | 390 ms | Success |
| 5       | Skill Search: "Java"    | 1,000 | 510 ms | Success |
| 6       | All Records Load        | 1,000 | 460 ms | Success |
| 7       | Dept: IT, Batch: 2024   | 1,000 | 430 ms | Success |
| 8       | Unverified User Filter  | 1,000 | 380 ms | Success |
| 9       | Event Feed Fetch        | 1,000 | 440 ms | Success |
| 10      | Multi-Parameter Search  | 1,000 | 500 ms | Success |
| Average |                         |       | 442 ms | PASS    |

#### Appendix F – Ethics / Institutional Approval

- **Data Privacy Protocols:** The system implements explicit user consent checkboxes during profile registration, allowing users to toggle phone number and personal email visibility.
- **Institutional Approval:** System authorization and approval granted under CSE Department Project Guidelines for the course CSA4305 – Internet Programming at SIMATS Engineering, Chennai.

#### Appendix G – Project Meeting / Progress Records

| Review / Date        | Agenda / Milestone                | Work Verified                   | Supervisor Feedback                          |
|----------------------|-----------------------------------|---------------------------------|----------------------------------------------|
| Review 1 (Week 3)    | Problem Statement & Schema Design | Requirement Specs & ER Diagram  | Approved ER layout with indexed search keys. |
| Review 2 (Week 8)    | Module Development & Frontend     | Authentication & Directory UI   | Recommended adding admin verification queue. |
| Review 3 (Week 13)   | Security Testing & Integration    | OWASP Checks & Taint Analysis   | Verified SQL injection mitigation.           |
| Final Viva (Week 16) | System Demonstration              | Live Prototype & Project Report | System passed all verification tests.        |

#### Appendix H – User / Industry Feedback

A User Acceptance Testing (UAT) survey was conducted with 15 participating alumni and students:

- **System Usability Score (SUS):** 86.5 / 100 (Graded "Excellent").
- **Key Feedback:** Users highlighted the directory filtering capabilities and clear distinction between verified and unverified profiles as major improvements over manual social network tracking.

#### **Appendix I – Publications / Patents / Competitions / Product Outcomes**

- **Product Prototype Outcome: AlumniConnect Web Application v1.0** deployed on local server environments for institutional demonstration.
- **Academic Submission:** Submitted as part of the partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering.

#### **Appendix J – Individual Contribution Evidence**

| <b>Candidate Name &amp; Reg. No.</b> | <b>Assigned System Modules</b>    | <b>Development Evidence</b>                                                                                                                                                               | <b>Contribution (%)</b> |
|--------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Ch.Sathish (192311032)               | Networking & Communication Module | Designed complete database, created frontend/backend interfaces, implemented OWASP security sanitization, executed query benchmarks, and authored complete capstone report documentation. | <b>100%</b>             |